Sahaj Bhadja

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Summary

Backend Software Engineer pursuing an MSCS at USC with 2+ years building and operating production data ingestion pipelines, APIs, and third-party integrations on PostgreSQL and cloud platforms, with strong observability, AI-assisted development, and interest in energy systems and grid modernization.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **data ingestion pipelines and event-driven async orchestration** on PostgreSQL with schema design and SQL tuning for 500+ users, owning services from design through production.
- Implemented **validation, reconciliation, and monitoring** across 5K+ monthly jobs with CI/CD releases and failure isolation, maintaining 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Designed and shipped **backend services and REST APIs** for 150+ users handling 10K+ daily requests, owning clean, testable features from design through deployment.
- Used **AI-assisted development tools** (Cursor, Claude) to accelerate coding, testing, and debugging, increasing delivery speed while sustaining stable performance under load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Built **large-scale data integrations and ETL ingestion** (AWS Glue, S3 object storage) with OOP services and 20+ test suites, improving response times **30%**.
- Automated **CI/CD deploys and integration testing** across three platforms, owning release pipelines in Agile sprints and boosting efficiency **68%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **monitoring, alerting, and logging** (Sentry, Azure Monitor, Kibana) for a microservice platform syncing 2M+ records, enabling incident response and continuous operation.
- Developed **REST APIs and third-party integrations** across eight services with Docker/Jenkins CI/CD, cutting deployment time **50%** on Azure.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Flask backend APIs and third-party integrations** with idempotent webhook handling, retries, and token-based auth across 50K+ records at sub-200ms latency.
- Engineered **time-series analytics pipelines** (Databricks, Spark) ingesting event data in hourly micro-batches, performing root cause analysis to cut manual handling **70%**.

Skills

Languages: Python, TypeScript/JavaScript, Java, SQL, C++, Bash
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, Webhooks, Microservices
Data & Pipelines: Data Ingestion, ETL, Time-Series Data, Event-Driven Systems, Message Queues, Idempotency & Retries
Databases: PostgreSQL, SQL Server, MySQL, MongoDB, Schema Design, Query Optimization
Cloud & DevOps: AWS (S3, Lambda, Glue), GCP, Serverless, Docker, Kubernetes, CI/CD
Observability & Tools: Monitoring, Alerting, Logging (Sentry, Kibana), Incident Response, AI Coding Tools (Cursor), Agile

Projects

FinNexus Live | *Plaid, Webhooks, Kubernetes, MongoDB*
- Built a **third-party integration platform** (Plaid, webhooks) on Kubernetes serving 50K+ users and 1M+ transactions with real-time updates and reliable, low-latency data serving.

Big-Brain Live | *GCP, Object Storage, PostgreSQL, REST APIs*

- Built a **data ingestion and indexing pipeline** on GCP with object storage processing 50K+ records, incremental updates, and tuning that cut latency **40%**.

Linguistix | *Python, AWS (ECS/Fargate), Docker, CI/CD*

- Developed a **containerized backend API** on AWS (ECS/Fargate) with automated CI/CD, validation checks, and 87% accuracy at production-grade reliability.